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The Einstein-Szilard Refrigerator: An Experimental Exploration

Keng Wai Chan, PhD, CEng **Malcolm McCulloch, PhD**

ABSTRACT

The eighty-year-old single pressure absorption refrigeration system invented by Albert Einstein and Leo Szilard is attractive as it has no mechanical moving parts and can be driven by heat alone. The literature on the refrigeration system is scarce and only theoretical analyses are available. In this paper adversities of the primitive design and the operation of the refrigerator prototype are discussed. The prototype is able to achieve temperatures between 8°C (46.4°F) and 14°C (57.2°F). Five pairs of heat inputs between 53 W and 111 W are studied. Cooling capacities decrease (from 34 W to 15 W) when the generator heat input increases, however, cooling capacities increase (from 17 W to 30 W) when the bubble pump heat input increases. The highest COP is 0.25 when the heat inputs to generator is 53 W and to bubble pump is 89 W. The ideal cooling capacity and COP are 40 W and 0.28, respectively.

INTRODUCTION

The Einstein-Szilard refrigerator has caught increasing attention in the last few years across the globe (Jha 2008; TIME 2008; BBC 2010; Worms 2010). This refrigeration system was invented by Albert Einstein and Leo Szilard eight decades ago (Einstein and Szilard 1928, 1930). The unique relationship among the three working fluids, ammonia, butane and water, creates a partial pressure gradient for the refrigerant to evaporate and cool the surroundings. The system involves no moving parts, hence it has the potential to be reliable. However, the design has hitherto never been commercially produced and little information about the patent is available (Dannen 1997). From 1930 to 1980, almost no work was done on this refrigeration system. Follin et al. (1980a, 1980b) intended to operate the refrigerator by using modest source temperatures (65°C [149°F] or below), which could be harvested from geothermal resources, solar thermal collectors, or cogeneration systems. In their study, the theoretical coefficient of performance (COP) of the system is 0.25 and 0.20, respectively. However, no further references or works pertaining to these studies are found.

An Einstein-Szilard refrigeration system without bubble pump was analyzed by Delano (1997) using a model derived from the principles of mass and energy conservation. The COP of the system is 0.35. Later on, Delano (1998) applied the Patel-Teja equation of state to assess the energy content of the substances and mixtures in each component of the complete refrigeration system. He derived an analytical model to estimate the mass flow rate of the bubble pump. For a 4 bar (58.0 psi) system, where temperatures of the evaporator, condenser and generator were –7°C (19.4°F), 43°C (109.4°F), and 102°C (215.6°F), respectively, the theoretical COP of the system was 0.17. Towards the end of his study, he built a working prototype of the refrigeration system. This prototype operated with isobutane, ammonia, and water (Delano 1998). Heat input to the generator and bubble pump of the prototype were 150–250 W and 50–70 W, respectively. The evaporator temperatures dropped to –2°C (28.4°F) when the condenser temperature was maintained at 21°C (69.8°F) using tap water. Nevertheless, there are no further results, information, or publications on this prototype. Shelton et al. (1999a) published two conference papers with Delano's findings. In the first paper, they presented the design analysis of the refrigeration system where they introduce a prototype, but only the working conditions of the prototype were mentioned. In the second paper, they reported their studies on the refrigeration system using the second law of thermodynamics (Shelton et al. 1999b).

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Schaefer (2000) converted the Einstein refrigeration system into a heat pump. She estimated properties of the working fluids with equations of state and evaluated the performance of the system using the mass and energy conservation equations. Three temperature pairs of the condenser and evaporator were studied. The COPs for the studied pairs were 1.51, 1.88, and 1.76, respectively. Several triplets of working fluids were investigated. The triplet of water/ammonia/butane has the highest COP of 1.88 at 5.25 bar (76.1 psi). Despite the thorough analysis of the cycle, she did not build or test any prototype.

Mejbri et al. (2006) investigated the feasibility of the Einstein refrigeration system. They evaluated two modified configurations using Delano's analytical model. In addition, they applied Delano's vapor/liquid ratio to estimate the mass flow rates of the system. The theoretical COP of the system was 0.18. However, the vapor/liquid ratio in Delano (1998) was derived from an air-lift pump model and the model was verified by using water only. Furthermore, the ratio used by Delano was obtained from a 4 bar (58.0 psi) system rather than Mejbri's 5 bar (72.5 psi) system. As the pressure changes, the ratio will change accordingly due to the variation in boiling point.

A research group from University of Shanghai for Science and Technology, China has published seven Chinese journal papers related to the Einstein refrigeration system (Wang et al. 2007a, 2007b; Wang et al. 2008; Song et al. 2005) and bubble pump (Wang et al., 2008; Tang et al. 2009a, 2009b). Among these papers, two pertain to the design of the Einstein refrigeration system. Wang et al. (2007a) presented a method to assess the system. They created measurement parameters for the components on a LabVIEW platform to analyze the system at different conditions. They presented the design parameters for the evaporator, condenser, and bubble pump of the refrigeration system (Wang et al. 2007b). They mentioned 'the prototype' in their papers, but no further experimental work or images are found. In addition, a lot of information presented in these papers is very similar to Delano's studies.

The above describes all literatures that reported on the Einstein refrigeration system. The system has been studied analytically by several researchers across the globe, yet only one working prototype has been built. Hitherto, all of them analyzed the system theoretically. What no one has yet able to produce is the experimental analysis. In this paper, the adversities of the primitive design to set up a refrigerator prototype are presented. In addition, the operation and performance of the refrigeration system is investigated experimentally. Furthermore, the pressure and temperatures in the refrigeration system were simply fixed by researchers (Delano 1998; Mejbri et al. 2006) theoretically and without any proven experimental data. An empirical model is proposed and verified experimentally to replace the existing predictions.

ADVERSITIES OF THE REFRIGERATOR PRIMITIVE DESIGN

The primitive design of the Einstein-Szilard refrigerator (Einstein and Szilard 1928, 1930) (Figure 1) works at single pressure. It consists of five main components: evaporator, generator, condenser, container, and a bubble pump. Three working fluids are used in this cycle: butane (refrigerant), ammonia (inert gas), and water (absorbent). The ammonia vapor from the generator is bubbled in the evaporator to reduce the partial pressure of the butane. As a result, the butane evaporates and cools the chamber. When the vapor mixture of butane/ammonia enters the condenser, the ammonia is absorbed by the sprayed water. This will restore butane's partial pressure and cause the condensation of butane. The ammonia/water solution is siphoned into the generator. Heat is applied to the generator to separate the ammonia from water. The remaining weak-ammonia solution is then pumped into the container by a bubble pump. Heat exchangers are used in the cycle for precooling to improve its efficiency.

The missing pressure heads. As shown in Figure 1, the generator is located below the condenser to create a head (h_1), which is larger than the ammonia bubbling depth (h_2). This pressure head difference creates a potential energy that allows the ammonia vapor produced in the generator to be transported and bubbled in the evaporator. In addition, these positions also allow the bubble pump to be located above the generator due to the communicating vessels principle. The liquid level of the bubble pump will reach the liquid level of the condenser.

The pressure heads proposed above are difficult to achieve in practice. The solution always tries to achieve static

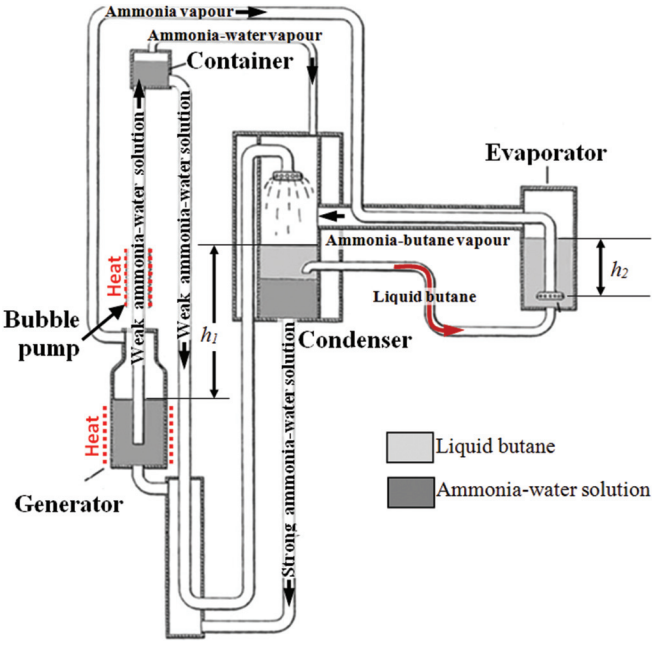


Figure 1 *Einstein-Szilard refrigerator (Einstein and Szilard 1928).*

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equilibrium. When the machine is stationary, the two tubes from the generator that connected to the container and to the evaporator have the same liquid level as the condenser due to the communicating vessels principle.

In Figure 1, water from the container is precooled by the counter flow from the condenser before entering the condenser. However, a small head increase in the container might be counterbalanced by the head loss in bends. This might increase system pressure as the absorption of ammonia vapor reduces due to the stop of water flow. As the pressure increases, the pumping rate of the bubble pump will decrease as more energy is needed to generate bubbles. For a system with fixed heat input to the bubble pump, the bubble pump might stop pumping if the energy is insufficient in producing bubbles to achieve the final void fraction for liquid lifting (Pfaff et al.1998). Eventually, the system may malfunction.

Bubble pump effect. When the generator starts producing ammonia vapor, the vapor enters the tube to the evaporator and creates the bubble pump effect (Figure 2). The ammonia vapor slugs carry water from the generator to the evaporator.

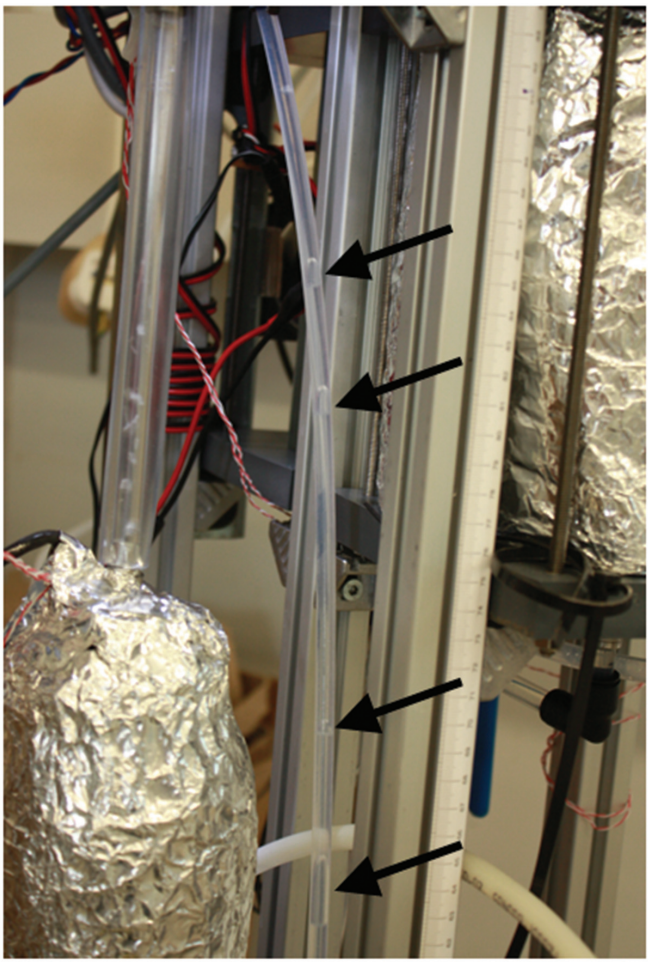


Figure 2 *The ammonia vapor in the tube to the evaporator from the generator creates a bubble pump effect.*

In addition, the generated ammonia vapor is reabsorbed by the water on its way to the evaporator. Mejbri et al. (2006) found that water in the evaporator will reduce the refrigeration system efficiency.

The return of butane. Figure 1 shows the flow direction of the liquid butane stream. The condensed liquid butane recirculates to the evaporator due to the communicating vessels principle. In the original design, the evaporator is situated next to the condenser at the level shown in Figure 1. This enables liquid butane to recirculate to the evaporator. Therefore, in order to transport the liquid butane from the condenser to the evaporator, a bent tube, similar to that shown in Figure 1, is used. This tube works well for liquid butane itself until water vapor from the generator condenses in the evaporator. Water with higher density displaces the liquid butane in the tube beneath the evaporator. Due to the density difference, the small liquid butane head change within the condenser is insufficient to force the liquid butane through the water layer. In other words, the water trapped in the tube blocks the flow of liquid butane from the condenser to the evaporator. If this condition continues, the system will collapse. As a result, the original design of the Einstein-Szilard single pressure absorption refrigeration system can hardly operate due to the physical constraints that exist in the system.

Delano's Prototype

Delano (1998) has built a working prototype of the Einstein refrigerator (Figure 3). The modifications he made to the refrigeration system have resolved the physical constraints of the system but at the same time have overlooked some of the good features in the original design. In other words, the modified

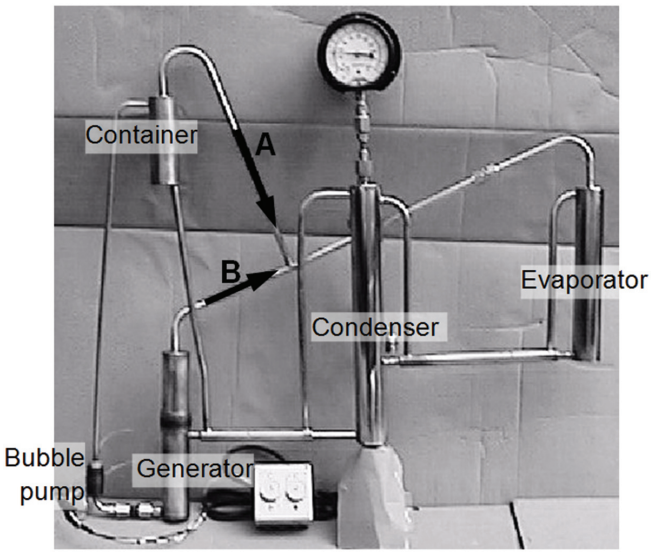


Figure 3 *The Einstein refrigerator prototype built by Delano (1998). The vapor stream from the container connected to the ammonia vapor stream from the generator.*

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system has to work harder. To avoid the bubble pump effect, Delano (1998) adjusted the height and position of the components and the connection. In his design, the position of the generator is elevated to give a clear passage to the ammonia vapor. The required pressure heads h_1 and h_2 are created on the spot while the generator starts operating. This pressure is essential to allow the ammonia vapor being transported to the evaporator.

In addition, Delano (1998) connects the vapor stream from the container (arrow A in Figure 3) to the ammonia vapor stream from the generator (arrow B in Figure 3), instead of connecting it to the condenser (Figure 1). There are advantages and disadvantages to doing so: the advantages are that the ammonia vapor is present in the container when the solution is heated in the bubble pump and also that the pressure built up in the container may further increase the ammonia vapor pressure from generator to increase the ammonia vapor bubbling in the evaporator. However, the water vapor present in the container may condense along the tube (arrow A). As the vapor condenses into water, it flows towards the generator due to gravity. These water droplets will encounter ammonia vapor from the generator (arrow B). The water will reabsorb the ammonia and reduce not only the amount of the ammonia vapor generated in the generator but also the efficiency of the system.

Furthermore, the height between the container and condenser was reduced compared to the original design. In Delano's design, the container is located exactly above the condenser. Water in the container creates a head for the flow to the condenser. However, as mentioned above, the water flow might reduce or stop as the head in the container is canceled out by the head loss in bends. This might lead to an increase in the system pressure and the stop of bubble pump.

As the generator is elevated to give a clear passage to the ammonia vapor, the bubble pump can no longer sit above the generator. This is because the water level cannot reach any level above the generator. Hence, the bubble pump has been relocated to be in parallel with the generator instead of in series with the generator. The parallel configuration yields a lower submergence ratio (H/L) compared to the series configuration, and it shrinks the performance of the bubble pump, where H and L denote the submergence height and length of the conveying tube, respectively.

In the prototype, Delano (1998) raised the position of the evaporator to allow the water that condenses in the evaporator to return to the condenser. However, the volume of liquid butane in the evaporator has been reduced significantly (compared with Einstein's original design) to avoid the generator being over filled with ammonia solution.

IMPROVED EINSTEIN-SZILARD REFRIGERATOR PROTOTYPE

The experimental rig (Figure 4) is the product of the Einstein's original system (Einstein and Szilard 1928, 1930) and Delano's modified system (1998). Compared to the original design, the generator in the new system is elevated to give clear passage to the ammonia vapor. At the same time, the bubble pump is shifted from the top to the side of the generator. These two modifications are similar to Delano's system. However, the vapor stream from the container to the condenser remains as in the original design. This is to avoid the weaknesses in Delano's designs. The chambers of the condenser and evaporator are made of transparent PVC, but the chambers of generator and container are made of aluminum tube due to their high operating temperatures. They are fixed on an aluminum frame and are connected using polytetrafluoroethylene (PTFE) tube. The heights of the container, generator, evaporator and condenser are 12, 20, 20, and 30 cm (4.72, 7.87, 7.87, and 11.81 in.), respectively.

The condenser, the container, the vapor stream from generator to evaporator and the water stream from container to condenser are precooled by using tap water through heat exchangers (as in Figure 5). As the condenser collects the heat from evaporator throughout the cooling process, water cooling of the condenser is essential to increases its heat dissipation rate. Because water with a lower temperature has a better ability to absorb ammonia, the container is cooled by tap water to

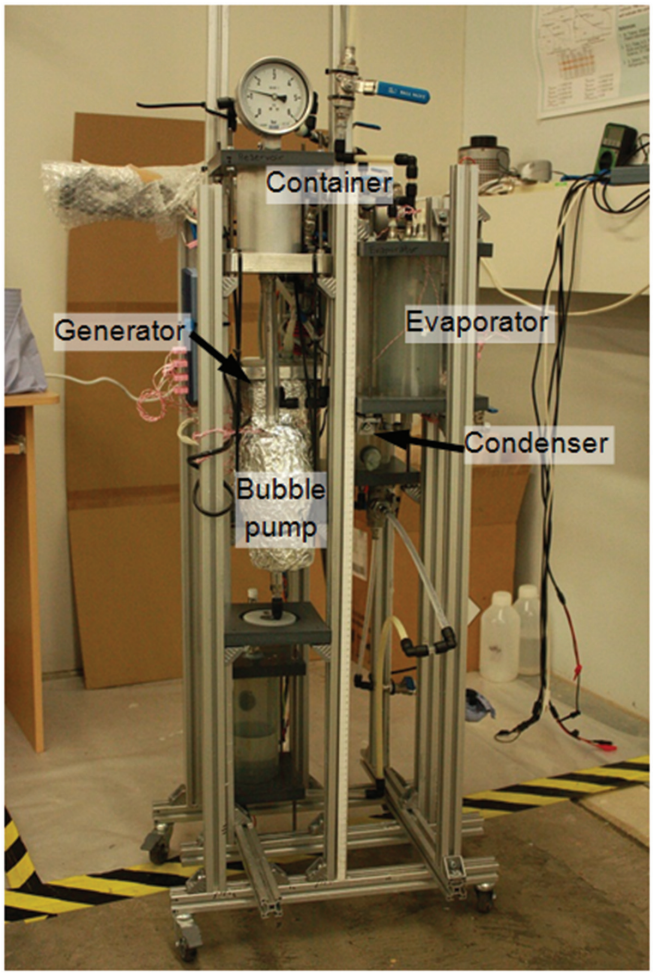


Figure 4 *The experimental setup.*

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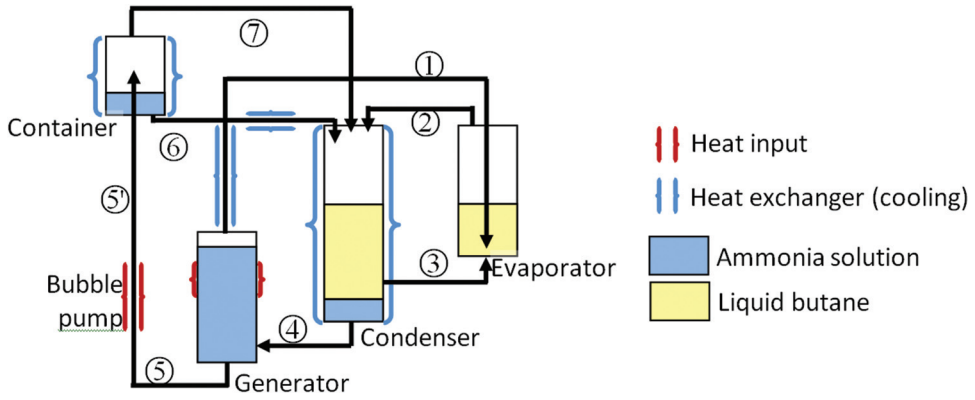


Figure 5 *The schematic diagram of the system.*

reduce the water temperature before it enters the condenser. The stream from generator to evaporator is also cooled by tap water to reduce the cooling burden of the evaporator. In order to reduce the head loss in bends, the stream from the container is cooled by tap water instead of the counter flow from condenser and it is fed into the condenser directly as shown in Figure 5.

All the chambers and heat exchangers are equipped with at least two N-type thermocouples (TCs), namely at the bottom and top of each component. All together, 24 TCs are logged through three sets of TC-08 eight-channel data loggers. In addition, the temperature distribution of the evaporator surface is observed by a Fluke Ti-10 thermal imager. It can capture not only the picture of the evaporator, but also the temperature distribution. Using the software provided, SmartView 3.1, the temperature distribution on the component surface can be extracted.

There are two components that require continuous supplies of heat in the Einstein refrigeration system, namely the bubble pump and the generator. In the bubble pump, heat is used to produce bubbles that carry water to a higher reservoir. Furthermore, heat is applied to the generator to separate the ammonia from the water. Therefore, two electric heaters are used in the system and are controlled by two 240 V variacs (autotransformers). Their energy use is logged using two multimeters through a PC interface.

The system is evacuated using a vacuum pump before it is charged with butane and ammonia solution. 1.04 kg (2.29 lb) of butane and 1.17 kg (2.58 lb) of ammonia solution (with the concentration of 25% ammonia [mass fraction] [Delano, 1998]) is charged into the system according to the dimension of the system in compliance with the two requirements: the ammonia solution in the generator should below the lid level (to avoid the bubble pump effect) and after charging the butane, the ammonia solution in the condenser should be below the evaporator (to allow condensed water in the evaporator to flow back to condenser).

OPERATION OF THE REFRIGERATION SYSTEM

The responses and reactions of the evaporator have behaved differently from conventional refrigerators. The temperature profile of the evaporator can be divided into three stages: heating, settling, and cooling (Figure 6). In the heating stage, when the system is supplied with heat, the temperature of the evaporator increases drastically reaching its maximum. Then, the temperature of the evaporator then drops to near its surrounding temperature in the settling stage. Finally, gradual cooling happens in the third stage.

After the experimental rig was left overnight (for 14 hours), the water absorbed the maximum possible amount of ammonia. When heat is applied to the generator, the ammonia is released from the solution due to its low boiling point (37.1°C [98.8°F] at 1 bar (14.5 psi), 56.7°C [134.1°F] at 2 bar (29 psi), 69.3°C [156.7°F] at 3 bar (43.5 psi), and 79.3°C [174.7°F] at 4 bar (58. 0psi) for 25% (mass fraction) ammonia solution [Conde-Petit 2006]). A minor cooling in the heating stage happens as the initial (cool) condenser able to absorb ammonia vapor that causes the liquid butane to evaporate due to a minor pressure decrease. However, when heat is continuously supplied to the generator, an abundance of ammonia is produced. The hot, excessive ammonia vapor bubbled in the evaporator not only raises the temperature of the evaporator but also raises the pressure of the system. At this stage, the amount of ammonia that is reabsorbed by the water is very small compared to the amount released. As a result, the concentration of the ammonia in the water within the generator reduces whilst the temperature of the evaporator and the pressure of the system reach their maximum for the first stage.

At the end of the first stage, the system has created two ideal conditions for the second stage: namely high pressure and low ammonia concentration in the water. In the second stage, the temperature of the evaporator drops due to the slow ammonia vapor bubbling whilst the pressure of the system drops due to the reabsorption of ammonia vapor. The temperature of the evaporator will drop close to the room temperature.

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The pressure comes to a local minimum when the absorption rate equals the generation rate at the end of the second stage.

In the third stage, as the ammonia vapor is bubbled in the evaporator steadily, the liquid butane evaporates and cools its surrounding gradually. The temperature of the condenser increases due to the heat absorption in the evaporator. As this temperature increases, the ammonia reabsorption reduces, and more ammonia remains in vapor form, hence increasing the pressure of the system.

PERFORMANCE OF THE SYSTEM

Thermodynamic models were used to analyze the system performance. Because the heat inputs to the bubble pump and the generator are known, the mass flow rates of the stream from the bubble pump ($\dot{m}_6$ and $\dot{m}_7$) can be predicted using equations presented by Delano (1998) and Chan and McCulloch (2013). The ammonia vapor stream from generator ($\dot{m}_1$) can be calculated by

$$\text{Mass flow rate, } \dot{m}_1 = \frac{\dot{Q}_{\text{gen}}}{h_{fg}} \tag{1}$$

The remaining mass flow rates ($\dot{m}_2$, $\dot{m}_3$, $\dot{m}_4$, and $\dot{m}_5$) could be calculated using the law of mass conservation (Delano, 1998). According to Delano (1998), the heat balance for the evaporator is expressed as (Patel and Teja 1982)

$$\dot{Q}_e + \dot{m}_3 h_3 + \dot{m}_1 h_1 = \dot{m}_2 h_2 \tag{2}$$

where h is computed using Patel-Teja equation of state.

In Figure 7, Process A is the heat dissipation process for refrigerant in the condenser. Process B is the refrigerant cooling process in the evaporator when the butane returns to the evaporator. As the partial pressure of the butane drops (induced by the inert gas), the refrigerant absorbs heat during its evaporation. It absorbs the sensible heat from the returned butane (Cooling 1), the inert gas (Cooling 2), and the surroundings (Cooling 3), as in Figure 7. Assuming pure ammonia gas at P_1 and T_1 enters the evaporator:

$$h_1 = h_{g,NH_3,P_1,T_1} \tag{3}$$

and pure liquid butane at P_3 and T_3 enters the evaporator

$$h_3 = h_{f,C_4H_{10},P_3,T_3} \tag{4}$$

For an ideal gas mixture of ammonia and butane at P_2 and T_2 exits the evaporator:

$$\begin{aligned} h_2 &= h_{g,C_4H_{10}+NH_3,P_2,T_2} \\ &= y_2(h_{g,C_4H_{10},y_2P_2,T_2}) + \\ &\quad (1-y_2)(h_{g,NH_3,(1-y_2)P_2,T_2}) \end{aligned} \tag{5}$$

where y_2 is the mole fraction of butane as it exits the evaporator.

Let P_1, P_2, and P_3 equal to system pressure, P_{sys}, and the partial pressure of butane, $P_b = y_2(P_{\text{sys}})$, Equation 2 becomes

$$\begin{aligned} \dot{Q}_e &= \dot{m}_3(h_{g,C_4H_{10},P_b,T_2} - h_{f,C_4H_{10},P_{\text{sys}},T_3}) \\ &\quad + \dot{m}_1(h_{g,NH_3,(P_{\text{sys}}-P_b),T_2} - h_{g,NH_3,P_{\text{sys}},T_1}) \end{aligned} \tag{6}$$

when $T_2 = T_{\text{sat},C_4H_{10},P_b}$, $T_3 = T_{\text{sat},C_4H_{10},P_{\text{sys}}}$, $h_{2,g,P_b} = h_{g,C_4H_{10},P_b,T_2}$, and $h_{2,f,P_{\text{sys}}} = h_{f,C_4H_{10},P_{\text{sys}},T_3}$, Equation 6 is written as

$$\begin{aligned} \dot{Q}_e &= \dot{m}_3(h_{2,g,P_b} - h_{3,f,P_{\text{sys}}}) \\ &\quad + \dot{m}_1(h_{g,NH_3,(P_{\text{sys}}-P_b),T_2} - h_{g,NH_3,P_{\text{sys}},T_1}) \end{aligned}$$

or

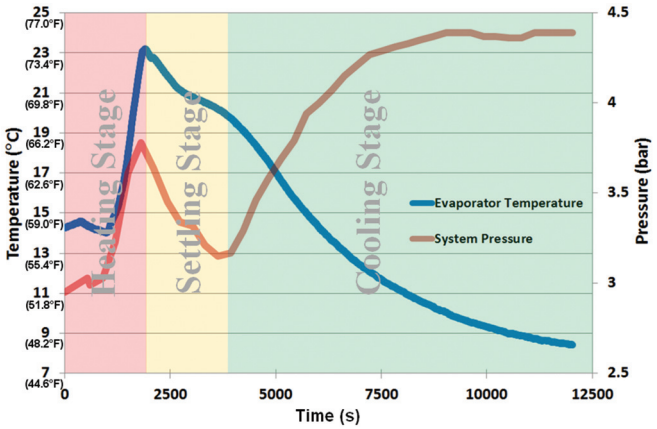


Figure 6 *The progression of evaporator temperature and system pressure.*

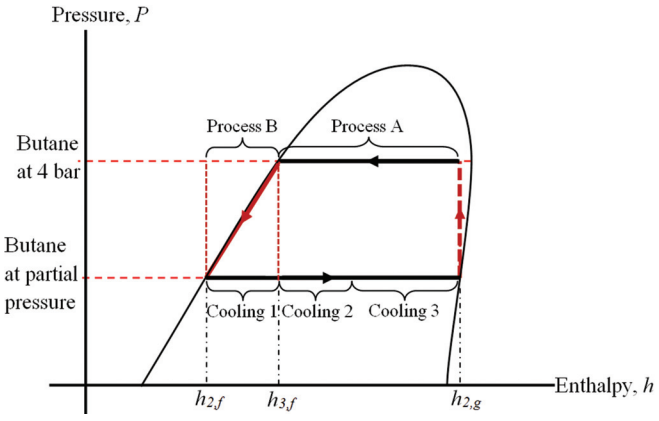


Figure 7 *Refrigerant (butane) on P-h diagram.*

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$$\dot{Q}_e = \dot{m}_3(h_{2,g,P_b} - h_{2,f,P_b}) + \dot{m}_3(h_{2,f,P_b} - h_{3,f,P_{sys}}) + \dot{m}_1(h_{g,NH_3,(P_{sys}-P_b),T_2} - h_{g,NH_3,P_{sys},T_1}) \quad (7)$$

where first term, second term, and third term are dedicated to the total heat absorbed by butane and the sensible heat dissipated by butane and ammonia, respectively. In the ideal case, when the butane and ammonia are precooled to the evaporator temperature (sensible heat of butane and ammonia is expelled) before entering the evaporator, the maximum ideal cooling capacity would be

$$\dot{Q}_{e,\text{ideal}} = \dot{m}_3(h_{2,g,P_b} - h_{2,f,P_b}) \quad (8)$$

As the heat input, system pressure and temperatures of each component were recorded, the mass and energy balance for the evaporator can be readily calculated.

RESULTS AND DISCUSSIONS

The two systems, namely the system with fixed heat input to the generator and the system with fixed heat input to the bubble pump, were tested. The voltage for both bubble pump and generator were between 75V and 95V, wherein for the first group of experiments the heat input to the bubble pump was set at 89 W (85V) and the heat input to the generator was varied between 53 W (75V) and 84 W (95V). In the second group of experiments, the heat power input to the generator was fixed at 67 W (85V) and the heat power input to the bubble pump was varied between 69 W (75V) and 111 W (95V). Throughout the experiments the evaporator was capable of reaching temperatures between 8°C and 14°C (46.4°F and 57.2°F) in a slow cooling process (Figure 8).

In the first study, as the heat input to bubble pump is fixed at 89W, the cooling capacity decreases from 34 W to 15 W when the heat input to the generator varies from 53 W to 84 W (Figure 9a). Meanwhile, the ideal cooling capacity decreases from 40 W to 25 W. Narayankhedkar & Maiya (1985) observed that the condensation rate of the condenser influences the cooling rate of the evaporator directly. If no condensation happens in the condenser, no cooling effect will take place in the evaporator. Hence, when more ammonia vapor is reabsorbed by water in the condenser, more liquid butane will evaporate in the evaporator and withdraw heat from its surrounding. Because the heat input to bubble pump is fixed, the amount of water that reaches the condenser is fixed. As the excessive ammonia vapor from the generator increases the system pressure, it reduces the partial pressure (of butane in evaporator) and induces the temperature decrease (Figure 9b). However, the cooling capacity and the refrigeration performance (COP) drop following the increase of system pressure. According to Patel-Teja equation of state (Patel and Teja 1982) and its application by Delano (1998) on this refrigeration

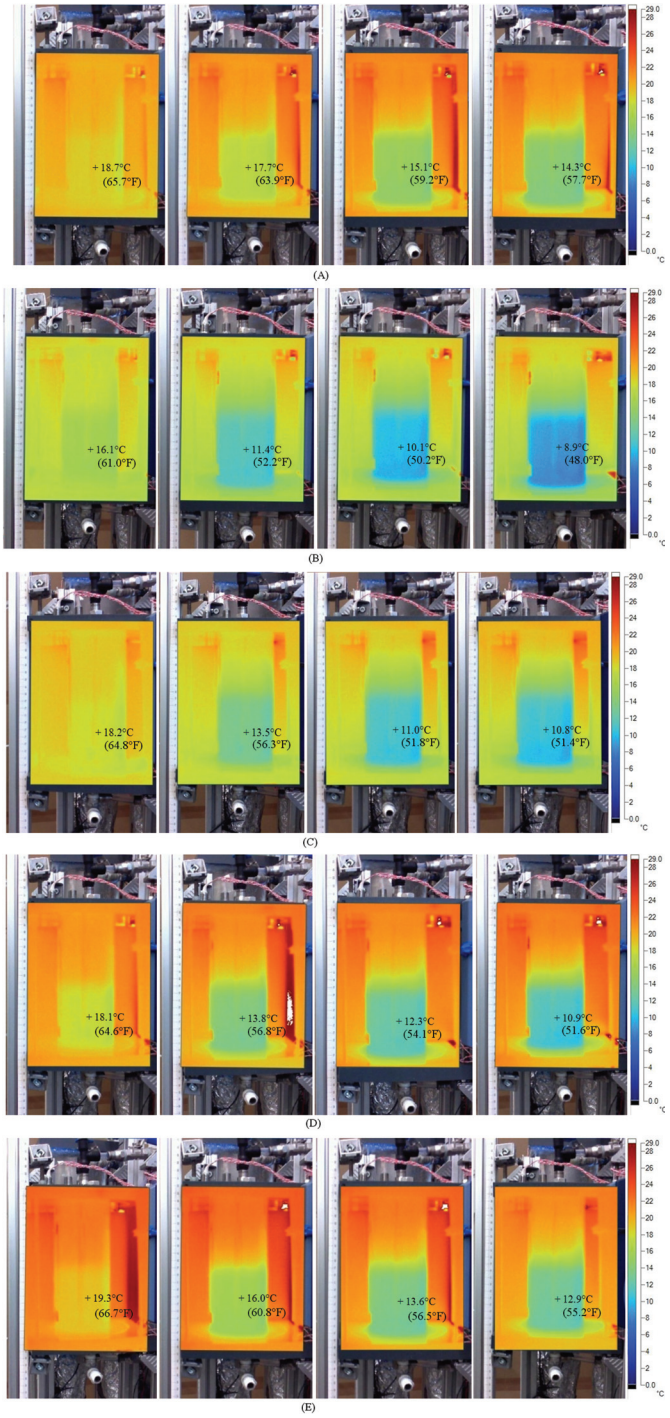


Figure 8 *Thermal image for the system: (a) at 75, 160, 280, and 345 minutes for 53 W (generator): 89 W (bubble pump); (b) at 70, 115, 160, and 225 minutes for 67 W: 89 W; (C) at 40, 70, 100, and 110 minutes for 84 W: 89 W; (d) at 55, 95, 135, and 195 minutes for 67 W: 69 W; and (e) at 90, 180, 270, and 310 minutes for 67 W:111 W.*

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system, the evaporation rate of butane in the evaporator reduces as the system pressure increases. When the heat input to the generator increases, the evaporator temperatures, COPs, and the ideal COPs for this study drop from 14°C to 8°C (57.2°F to 46.4°F), 0.25 to 0.08, and 0.28 to 0.15, respectively.

In the second study, as the heat input to the generator is fixed at 67 W, the cooling capacity varies between 17 W and 30 W when the heat inputs to the bubble pump are between 69 W and 111 W (Figure 10a. Meanwhile, the ideal cooling capacity increases from 25 W to 37 W.Because the heat input to the generator is fixed in this study, the production of ammonia vapor and the system pressure is fixed indirectly. The rise of heat input (to the bubble pump) increases the water volume to the condenser. This might increase the reabsorption of the ammonia and the condensation of butane in condenser. Hence, there might be a small system pressure drop to increase the ammonia vapor generation in the generator and the cooling capacity accordingly. However, the pressure gradient would be counterbalanced by the increase of the ammonia vapor generation in the generator. As shown in Figure 10b, the system pressure of this study did not change like the first study. When the heat input to the bubble pump increases, the temperature of the evaporator is between 8°C and 13°C (46.4°F and 55.4°F), COPs are between 0.10 and 0.15, and the ideal COPs are between 0.15 and 0.20, respectively.

CONCLUSION

The results of this experiment confirm that the primitive design of the Einstein-Szilard refrigeration system can operate with a few adjustments and modifications. Despite the slow cooling process, the first set experimental data of the Einstein-Szilard refrigeration system has been obtained. The evaporator is capable of reaching temperatures between 8°C and 14°C (46.4°F and 57.2°F). For the first study, heat input to bubble pump was fixed at 89 W and the heat input to generator varied between 53 and 84 W. More ammonia vapor is generated as the heat input rises, causing the system pressure to increase and reduce the evaporation of butane (in evaporator). Thus, the

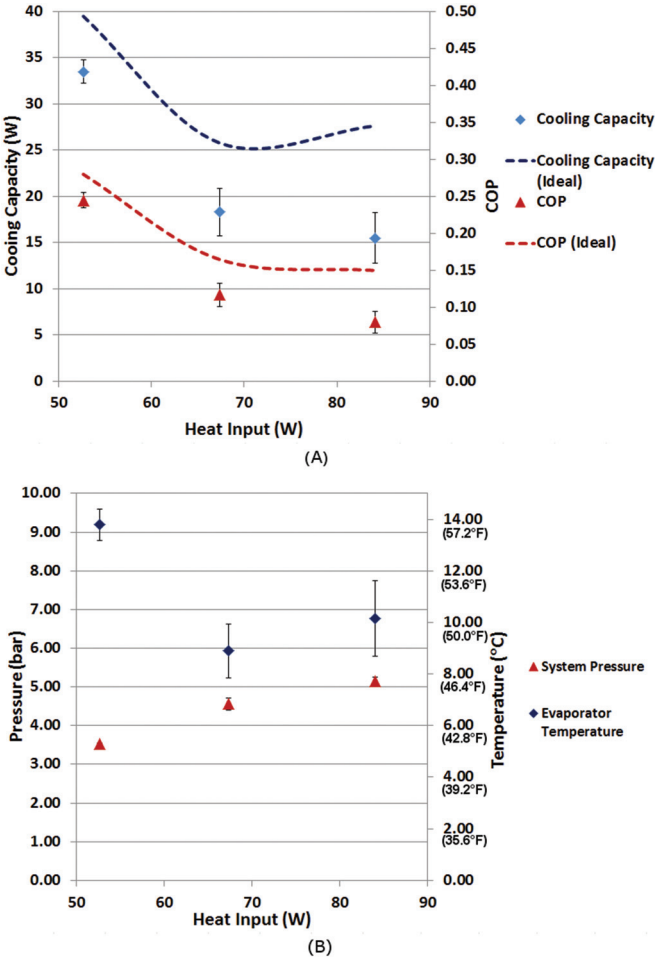


Figure 9 *Performance of the system with fixed bubble pump heat input: (a) cooling capacity and COP and (b) system pressure and evaporator temperature.*

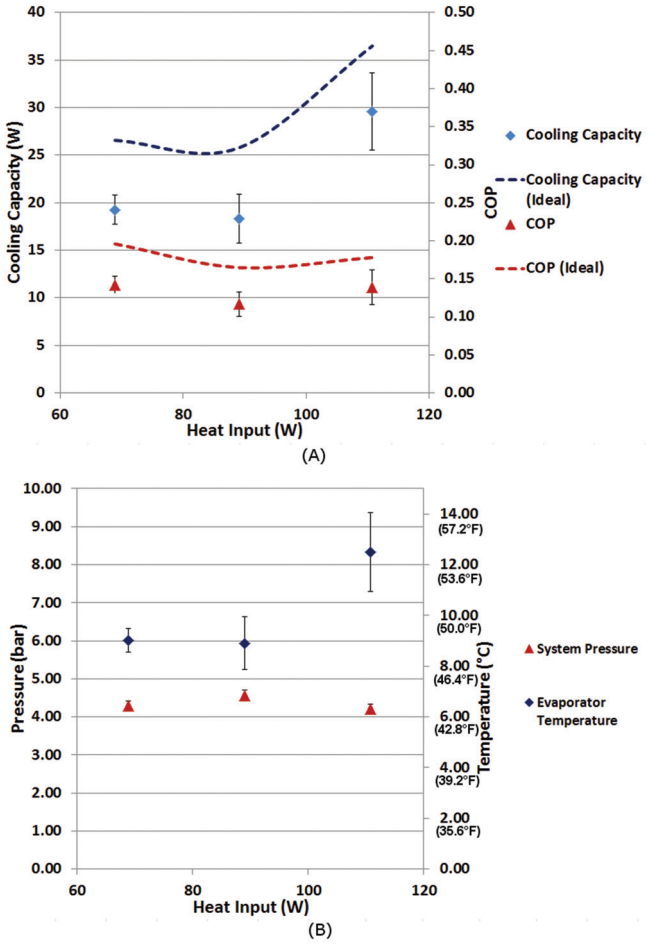


Figure 10 *Performance of the system with fixed generator heat input: (a) cooling capacity and COP and (b) system pressure and evaporator temperature.*

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cooling capacities for this study drop from 34 to 15 W when the heat input to generator increases. Meanwhile, heat input to generator was fixed at 67 W and the heat input to the bubble pump was between 69 and 111 W for the second study. The system pressure of this study did not change like the first study. The rise of heat input increases the volume of water to the condenser and causes the reabsorption of the ammonia and the condensation of butane in the condenser to increase. This might induce a small pressure drop and cause the ammonia vapor generation (in the generator) and the cooling capacity to increase accordingly. For the second study, the cooling capacities increase from 17 to 30 W as the heat input to bubble pump increases. The best experimental COP among these systems is 0.25 when the heat inputs to generator and bubble pump are 53 and 89 W, respectively. The performance of the system could be improved when the butane and ammonia are precooled to the evaporator temperature (sensible heat of butane and ammonia is expelled) before they enter the evaporator. The ideal cooling capacity and COPs for the first study could therefore increase to between 40 and 25 W and 0.28 and 0.15, respectively. Meanwhile the ideal cooling capacity and COPs for the second study are between 25 and 37 W and 0.15 and 0.20, respectively.

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NOMENCLATURE

H	=	submergence height, m (in.)
L	=	length of conveying tube, m (in.)
P_{sys}	=	system pressure, bar (psi)
P_b	=	partial pressure of butane, bar (psi)
$\dot{Q}_{bp}$	=	heat input to bubble pump, W (Btu/h)
$\dot{Q}_e$	=	cooling capacity, W (Btu/h)
$\dot{Q}_{gen}$	=	heat input to generator, W (Btu/h)
T_{gen}	=	generator temperature, °C (°F)
h_f	=	enthalpy of saturated liquid, kJ kg^{-1} (Btu lb^{-1})
h_g	=	enthalpy of saturated vapor, kJ kg^{-1} (Btu lb^{-1})
h_{fg}	=	enthalpy of evaporation, kJ kg^{-1} (Btu lb^{-1})
$\dot{m}$	=	mass flow rate, kg s^{-1} (lb s^{-1})

REFERENCES

BBC. 2010. Einstein's Fridge. BBC's Wallace and Gromit—World of Invention. 17th Nov 2010.

Chan, K.W. and M. McCulloch. 2013. Analysis and modeling of water based bubble pump at atmospheric pressure. *International Journal of Refrigeration* 36:1521– 28.

Conde-Petit, M.R.,2010. Thermophysical Properties of NH3 + H2O mixtures for the industrial design of absorption refrigeration equipment. Zurich, Switzerland: M. Conde Engineering.

Dannen, G. 1997. Einstein-Szilard refrigerators. *Scientific American* (January):90 – 95.

Delano, A. 1997. Analysis of the Einstein refrigeration cycle. Master's thesis, Georgia Institute of Technology.

Delano, A. 1998. Design analysis of the Einstein refrigeration cycle. PhD thesis, Georgia Institute of Technology.

Einstein, A., and L. Szilard. 1928. Improvements relating to Refrigeration Apparatus. GB Patent 282428.

Einstein, A., and L. Szilard. 1930. Refrigeration. US Patent 1781541.

Follin, J.W., and K. Yu. 1980a. Energy conversion and storage techniques: evaluating the Einstein refrigerator. *Energy Programs—Quarterly Report*. Johns Hopkins University Applied Physics Laboratory, July–September.

Follin, J.W., P. P. Panddfini, and K. Yu. 1980b. Energy conversion and storage techniques: Evaluating the Einstein refrigerator. *Energy Programs—Quarterly Report*. Johns Hopkins University Applied Physics Laboratory, October–December.

Jha, A. 2008. Einstein fridge design can help global cooling. *Guardians* 21st September.

Mejbri, K., N.B. Ezzine, Y. Guizani, and A. Bellagi. 2006. Discussion of the feasibility of the Einstein refrigeration cycle. *International Journal of Refrigeration* (1):60–70.

Narayankhedkar, K., and M. Maiya. 1985. Investigations on triple fluid vapour absorption refrigerator. *International Journal of Refrigeration* 335–42.

NIST. 2008. Thermophysical properties of fluid systems. National Institute of Standards and Technology. http://webbook.nist.gov/chemistry/fluid/.

Patel, N.C., and A.S. Teja. 1982. A new cubic equation of state for fluids and fluid mixtures. *Chemical Engineering Science* 37, 463–73.

Pfaff, M., R. Saravanan, M. Prakash Maiya, and S. Srinivasa Murthy. 1998. Studies on bubble pump for a water-lithium bromide vapor absorption refrigerator. *International Journal of Refrigeration* 21:452–62.

Schaefer, L.A. 2000. Single pressure absorption heat pump analysis. PhD Thesis, Georgia Institute of Technology.

Shelton, S.V., A. Delano, and L.A. Schaefer. 1999a. Design analysis of the Einstein refrigeration cycle. *Proceeding, Renewable and Advanced Energy Systems for the 21st Century*, RAES-4.

Shelton, S.V., A. Delano, and L.A. Schaefer. 1999b. Second law study of the Einstein refrigeration cycle. *Proceeding, Renewable and Advanced Energy Systems for the 21st Century*, SLA-2.

Song, M.F., D.P. Liu, and W.J. Huang. 2005. Development of single pressure absorption-type refrigeration technology. HVAC 35:31–35.

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Tang, C.W., D.P. Liu, Y.X. Qi, and R.J. Wang. 2009a. Performance of bubble pump in single-pressure Einstein absorption refrigerator. J. Ref. 30:35 – 39.

Tang, C.W., Y.X. Qi, D.P. Liu, W.J. Li, Y.Q. Ping, and X.M. Xue. 2009b. Experimental investigation on bubble pump in Einstein refrigeration cycle. Cryo. & Supercond: 37:55–59.

TIME. 2008. Best inventions of 2008: Einstein's fridge. *Time* 172(19):41–64.

Wang, R.J., and D.P. Liu. 2008. Exergy analysis of improved Einstein refrigeration cycle. *Journal of Industrial and Engineering Chemistry* (China) 59:820–24.

Wang, R.J., D. P. Liu, and X.M. Xue. 2007a. Design of the single pressure absorption refrigerator with Einstein cycle. Machinery Design & Manuf., 12, 37 – 39.

Wang, R.J., D. P. Liu, and X.M. Xue. 2007b. Parameter measurement of improved Einstein cycle refrigerator. Dev. & Inn. Machinery & Electr. Product 20:127–28.

Wang, R.J., D.P. Liu, X.M. Xue, and D.L. Zhong. 2008. Parameter design and determination for bubble pump in Einstein absorption refrigerator. *Fluid Machinery* 36:62–65.

Worms, P. 2010. *Wallace and Gromit's World Of Invention*. New York: HarperCollins.

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